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Artificial intelligence enabled cyber security defense for smart cities: A novel attack detection framework based on the MDATA model

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ABSTRACT

Smart cities have attracted a lot of attention from interdisciplinary research, and plenty of artificial intelligence based solutions have been proposed. However, cyber security has always been a serious problem, and it is becoming more and more severe in smart cities. The existing attack defense methods are not suitable for detecting multi-step attacks since the detection rules are limited and the efficiency is limited by a large number of false security alarms. Hence, an advanced solution is urgently needed to improve cyber security defense capability. In this paper, we propose a novel attack detection framework called ACAM. To better represent the cyber security knowledge, the framework is based on the MDATA model, which can represent dynamic and temporal–spatial knowledge better than the knowledge graph. The framework consists of the knowledge extraction module, the subgraph generation module, the alarm correlation module, and the attack detection module. These modules can remove false alarms and improve the detection capabilities of multi-step attacks. We implement the framework and conduct experiments on the cyber range platform, the experimental results validate the good performance of attack detection accuracy and efficiency. The framework can greatly improve the cyber security defense capabilities for smart cities.

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1. Introduction

With the rapid development of information technologies, smart city projects have been attracting more and more attention around the world. For example, the cloud-based Internet of Things (IoT) is making the manufacturing process more intelligent [1,2], the next-generation mobile communication technologies are making transportation smarter [3,4], and Artificial Intelligence (AI) based diagnostic technologies are making healthcare more convenient [5]. However, the cyber security problem always exists behind information technologies due to unavoidable technological vulnerabilities and weaknesses [6]. With more smart solutions deployed in smart cities, the cyber security problem becomes more serious. Traditional cyber attacks may probe the system's vulnerabilities and exploit them to invade the system, while in smart cities, any weakness might be used to invade the whole system. For example, the attacker can invade the cloud server through a fragile IoT device, and control the system through illegal access [6,7]. Hence, a novel cyber security defense method is urgently needed in smart cities.

Mainstream cyber security defense methods rely on attack detection rules [8]. Many cyber security products can produce security alarms according to the detection rules, such as Intrusion Detection Systems (IDS), firewalls, etc. However, with the fast development of cyber attack technologies, it becomes more and more difficult to detect since many of the detection rules are out-of-date or the attackers choose to split the attack steps in order to evade detection. With the emphasis on cyber security, more and more institutes and enterprises study the latest attacks and share the reports for detection. Some methods extract the detection rules automatically [9,10] and detect the attacks according to the latest detection rules. However, most methods only focus on the detection of single-step attacks with high accuracy, and only a few methods detect multi-step attacks. The multi-step attack detection methods combine and correlate the security alarms by the detection rules, but the huge number of security alarms in smart cities will increase the detection time dramatically. Therefore, a novel and efficient cyber attack detection method is critical and in demand, which can largely improve cyber security defense capabilities.

In this paper, we study AI enabled cyber security defense for smart cities. Recently, some deep learning based and knowledge graph based cyber attack detection methods have been proposed [11–13], while deep learning models lack theoretical explanation, and many adversarial attacks have revealed

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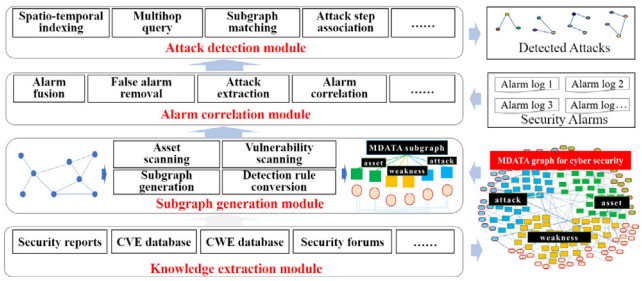

Fig. 1. An overview of the ACAM framework.

the weaknesses of models [14–16]. In contrast, knowledge graph based models can explain the results clearly, but there are some intrinsic problems in the cyber security area. First, cyber security knowledge changes quickly such as attack strategies and attack steps. Second, the alarms and logs cannot be represented efficiently due to the temporal and spatial characteristics, while they are very critical for attack detection. Therefore, we select a MDATA (Multi-dimensional Data Association and inTelligent Analysis) model [17], a novel knowledge representation model that can represent dynamic and temporal–spatial knowledge to enable cyber attack detection.

Specifically, we propose an Advanced Cyber security Analysis framework based on the MDATA model (ACAM for short). The intuitive idea is to convert the cyber security data to the MDATA graph and use the graph matching technique to detect multi-step attacks. The ACAM framework contains the following modules as depicted in Fig. 1. *(1) The knowledge extraction module* extracts different kinds of cyber security knowledge and converts them into large-scale MDATA graph, including the relations between assets, vulnerabilities, and attack steps. *(2) The subgraph generation module* generates a subgraph according to the system topology and assets. The generated subgraph contains possible detected attacks, which can reduce many unrelated security alarms to improve detection efficiency. *(3) The alarm correlation module* converts the security alarms to a real-time MDATA graph and correlates security alarms with attacks. *(4) The attack detection module* applies the subgraph matching techniques to detect the multi-step attacks that adhere to the spatial–temporal constraints. These modules enable efficient attack detection by removing false alarms and accelerating detection by graph computing.

We implement the attack detection framework on the cyber range platform and extensively evaluate the proposed method. As the MDATA model is applied in our framework to support attack detection, we first evaluate the knowledge management efficiency, and the results show that the multi-hop query time is much less than other graph computing systems. Furthermore, we conduct extensive attacks on a virtual network on the cyber range platform to evaluate detection accuracy and efficiency. The results show that the average detection accuracy of different types of attacks can reach 90.6%, and the average detection delay is as low as 14 ms.

The contributions of the paper are summarized as follows.

- To the best of our knowledge, this is the first work in the area of cyber security to build a cyber attack detection framework based on the MDATA model, which significantly improves the detection performance;
- The proposed ACAM framework can detect cyber attacks that consist of multi-steps. By removing many false alarms, the detection efficiency is greatly improved;
- We conduct extensive experiments on the cyber range platform to evaluate the cyber attack detection performance, and the results confirm the performance.

The remainder of the paper is organized as follows. The next section highlights the related works on cyber attack detection. Preliminaries are provided in Section 3. We describe the ACAM framework for detecting multi-step attacks in detail in Section 4. We show the experimental results and analysis of the cyber range platform in Section 5. Finally, we conclude the paper in Section 6.

2. Related work

In this section, we introduce related works from three aspects: cyber attack detection model, single-step attack detection and multi-step attack detection.

2.1. Cyber attack detection model

The attack tree model [18] uses the tree structure to describe the process of cyber attack, which can intuitively represent the possible risks of the system. The attack graph model [19] describes the attack path graph of the attacker in the process, including all attempted attack paths. The model can describe the relationship between vulnerability in the network and information systems. The kill chain model [20] is proposed to represent the attack path taken by attackers to invade information systems. The attack surface model [21] is used to describe the security performance of the system from the perspective of the system's own resources. Compared to these models, such as the attack graph, the attack surface model focuses on examining all vulnerabilities at the entry and exit points of the system. The Mitre company proposed the ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge) model[1] to abstract related attack tactics and techniques from the perspective of attackers. Hence, it is regarded as a high-level abstract model of the attacker's lifecycle. Compared to the kill chain model, the ATT&CK model provides more fine-grained and easily shareable knowledge.

2.2. Single-step attack detection

Single-step attack detection, also known as single-step intrusion detection, for supporting multi-step attack detection. The detection methods can be divided into rule-based detection methods and anomaly-based detection methods.

The rule-based (signature) detection methods use the existing attack detection rules and pattern matching against known attacks to determine whether an attack has occurred, such as Snort.[2] With the discovery and spread of vulnerabilities, existing detection rules need to be continuously updated and improved to prevent intrusion by attackers. Manually generating detection rules for known attacks is time-consuming and error-prone. Therefore, some researchers propose using an automated way to generate detection rules for known attacks. An automatic detection rules generation method for Suricata[3] is proposed in [9] through deep packet inspection techniques. A Bayesian method is proposed in [10] to automatically generate IF–THEN detection rules. Cyberspace is complex and changing, *0*-day and *n*-day attacks are used frequently, which severely limits the capabilities of rule-based detection methods.

Anomaly-based detection methods model known normal behavior patterns, and any deviation from the model is considered an anomaly. The method generates a large number of false alarms because it is impossible to know all normal behavior patterns. As

[1] https://attack.mitre.org
[2] https://www.snort.org
[3] https://suricata-ids.org

the data became available, the researchers explored the possibility of single-step attack detection using machine learning techniques for more effective attack detection. The pipeline of these methods includes feature engineering and model classification, where the features are designed by experts [22]. Deep learning methods can automatically extract complex features from raw data without feature engineering. For example, a deep learning-based method is proposed in [23] to convert raw network traffic data into a hierarchical structure and then use convolutional neural networks to detect attacks. These methods can achieve very high attack detection accuracy when relying on a large amount of available data, but cannot cope with unknown attack detection. Furthermore, FC-Net is proposed in [24] to explore the problem of unknown attack detection in few-shot scenarios.

As cyber attack techniques become more and more complex, single-step attack detection methods cannot effectively meet the needs of attack detection due to the lack of context.

2.3. Multi-step attack detection

A multi-step attack generally refers to a set of attack steps taken by an attacker for a specific target, and the set of attack steps includes at least two different attack behaviors. To address the problem of multi-step attack detection, researchers have proposed many methods, which can be classified into correlation-based techniques and provenance graph-based techniques.

The core of correlation-based multi-step attack detection is to use multi-dimensional information comprehensive analysis to find out attack steps, form the attack path, and finally reveal the attacker's intention. Correlation-based multi-step attack detection has been widely studied, including various security defense measures and analysis techniques, such as alarm generation, correlation analysis, and attack path discovery. Attack path discovery is also referred to as scene reconstruction in some works. The data sources used for analysis are alarms generated by single-step attack detection, such as system audit logs or low-dimensional code information. Some researchers use similarity to measure attack alarms [25]. In [26], the similarity relationship between alarms is described by Equality Constraints Sets. From the anomaly detection perspective, [27] leverages normal user behaviors to model statistical relations for anomaly detection. The approach is prone to false positives since abnormal behavior does not represent an attack event. Some methods increase the causality between multi-step attacks and have better interpretability [28]. Hidden Markov Models (HMM) are used to define specific stages in a multi-step attack, and probabilities are then used to calculate the probability that a security event belongs to each attack stage. [29]. A new HMM method considering the problem of interleaved multi-step attacks is proposed in [30]. Other methods consider structural information in the asset dimension from the perspective of the defender. Correlate alarms and attack graphs first proposed in [31]. Furthermore, the principle of causal association combines information such as the topology of the network structure [32]. However, these methods are less versatile and require constant updating and maintenance of the asset dimension information. Similarly, a causal graph is proposed in [33] to leverage expert knowledge for real-time detection.

The core of multi-step attack detection based on provenance graphs is to use graph mining and analysis technology to discover the attacker's attack path. Existing research is mainly based on the system audit logs, which contain all information about interactions between the system user and the system kernel. BackTracker [34] was one of the first works to use provenance graphs to identify attack paths. In [35], an attack reconstruction method is proposed, which uses labels to perform multi-step attack detection on the provenance graph to meet the real-time requirements. In addition, a prototype system is implemented in [36] that adds probabilistic bayesian networks to provenance graphs to identify zero-day attacks. Provenance graphs with kill chains are combined in [37] to map low-level system audit log information to a high-level representation. Tactical provenance graphs are introduced in [11] to describe the lowest-level attack pattern by combining endpoint detection and provenance graphs. Observing that different attacks may share similar abstract attack strategies, ATLAS [38] combines natural language processing technology and provenance graphs to propose a sequence-based attack semantic model.

The machine learning-based methods are poorly interpretable and cannot respond effectively according to the results of model detection. The provenance graph-based methods rely on system audit logs, which are difficult to associate with multi-dimensional information for effective detection. Knowledge representation models, such as knowledge graphs, have reasonable explanations. However, the existing attack detection frameworks or methods based on knowledge representation models are poor in versatility and cannot meet the continuous dynamic update of assets, vulnerabilities, and attack knowledge.

3. Preliminaries

In this section, we introduce the definitions and notations in this paper. We also formally formulate the problem of multi-step attack detection.

3.1. System model

Consider a large networked system $G = (V, E)$, where each node $v_i \in V$ represents a node (such as a computer or server) in the system and an edge (v_i, v_j) represents two nodes v_i and v_j are connected. If two nodes are connected, it implies that the attacker can try to attack one node if another is invaded. For each node, denote the asset on node v_i as $A_G(v_i) = \{a_1(v_i), a_2(v_i), \ldots, a_n(v_i)\}$, where each $a_k(v_i) \in A_G(v_i)$ represents a specific asset such as versions of software or hardware. Since many attacks exploit the vulnerabilities or weaknesses of the system, we denote the set of vulnerabilities and weaknesses as $W_G(v_i) = \{w_1(v_i), w_2(v_i), \ldots, w_m(v_i)\}$ after vulnerability scanning.

An attacker can invade the networked system starting from any node in the system. Usually, the attacks are divided into many attack steps to evade detection, such as the APT (Advanced Persistent Threat) attacks. We assume that a multi-step attack on the system G is denoted as a sequence $S = \{s_1, s_2, \ldots, s_k\}$, where each step s_i is regarded as a specific attack step, such as active scanning, SQL (Structured Query Language) injection, etc. Normally, each attack step can be represented as a tuple:

$$s_i = (IP_s, Port_s, attack_j, IP_d, Port_d, time),$$

where IP_s, $Port_s$ denote the source IP and source port of the attack step, IP_d, $Port_d$ denote the destination IP and destination port, and $attack_j$ denotes the attack techniques.

For example, a typical multi-step attack is illustrated in Fig. 2, where 5 attack steps exist. The attacker (assume $IP_1 = 10.XX.XX.XX$) first uses SQL injection to invade a web server (assume $IP_2 = 11.1X.XX.XX$). Then the attacker uses active scanning to find the vulnerability of an important server (assume $IP_3 = 11.2X.XX.XX$), and uses password cracking to invade the server. In the fourth step, the attacker installs remote control tools on the server. Finally, sensitive files are transmitted to the attacker.

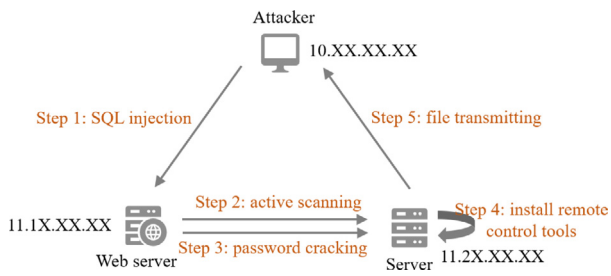

Fig. 2. An illustration of a multi-step attack process.

In the whole, the multi-step attack can be described as $S = \{s_1, s_2, \ldots, s_5\}$, where

$$\begin{cases} s_1 = (IP_1, port_1, SQL\ injection, IP_2, port_2, t_1) \\ s_2 = (IP_2, -, active\ scanning, IP_3, -, [t_2, t_3]) \\ s_3 = (IP_2, port_3, password\ cracking, IP_3, port_4, t_4) \\ s_4 = (IP_3, -, install\ remote\ control\ tools, -, -, t_5) \\ s_5 = (IP_3, port_5, file\ transmitting, IP_1, port_6, t_6) \end{cases} \quad (1)$$

In a networked system, the security products are deployed and can produce security alarms for a specific attack. For example, an IDS produces an alarm to the SQL injection attack and then stores it in the log file. We denote each alarm to a specific attack as $alarm_i$ when an attack $attack_i$ happens. Normally the $alarm_i$ is in contextual form, which contains enough information such as source IP, source port, destination IP, destination port, attack description, etc. However, a large number of security alarms are produced every day and many of these are false alarms, which makes the detection of multi-step attacks much more difficult.

3.2. Problem definition

In the networked system G, we can obtain the asset and vulnerability sets (A_G and W_G) in advance. The security products can produce alarms and we denote the set as $ALARM_G = \{alarm_G(t_1), alarm_G(t_2), \ldots\}$. The objective is to detect the attacks imposed on the system. We formulate the problem as follows.

Problem 1. The multi-step attack $S(1), S(2), \ldots$ The multi-step attack S is launched on the networked system G, where $S(i) = \{s_1(i), s_2(i), \ldots, s_k(i)\}$. The problem is to detect all attacks $S(i)$ and the corresponding detailed attack steps according to the security alarms $ALARM_G$.

4. ACAM: Advanced cyber security analysis framework based on the MDATA model

In this section, we describe the Advanced Cyber security Analysis framework based on the MDATA model (ACAM) for attack detection in detail. First, we present an overview of the framework and explain the role of the MDATA model. We then describe the modules in the framework and introduce some methods that are implemented. Finally, we discuss the advantages and disadvantages of the proposed framework.

4.1. Overview of the framework

The framework is illustrated in Fig. 1, which contains four main modules and a MDATA graph for cyber security.

- *The knowledge extraction module* is in charge of extracting cyber security knowledge from various kinds of security data sources, such as security reports, vulnerability databases, cyber threat intelligence, and news from the security forums. Furthermore, the module converts the extracted knowledge into a MDATA graph, which maintains a large-scale and comprehensive cyber security knowledge base.
- *The subgraph generation module* is introduced to generate a specific subgraph from the MDATA graph according to the network topology and asset distribution of a networked system. The generated subgraph can retain the possible attacks that may be launched to the networked system and thus can reduce the huge number of false alarms.
- *The alarm correlation module* is used to convert the real-time security alarms from the log file into a MDATA graph and associate the alarms with the corresponding attacks in the networked system. The generated graph based on the alarms is referred to as a real-time attack graph.
- *The attack detection module* is designed to detect multi-step attacks through graph computing techniques, such as subgraph matching and multi-hop querying.

The established MDATA graph for cyber security maintains the latest knowledge, including cyber attacks, assets and vulnerabilities. In addition, the MDATA graph can be easily updated when some knowledge varies, which shows superior performance compared to the knowledge graph. As the security alarms are produced constantly and sequentially, the proposed framework can establish and updates the real-time attack graph for detection as soon as a new alarm is generated. Hence, the framework can realize the persistent detection capability for multi-step attacks.

4.2. Knowledge extraction module

Cyber security knowledge continues to increase as attack technologies change or new vulnerabilities emerge. A large amount of relevant knowledge is released every day, and many organizations and institutions build a variety of knowledge bases to effectively manage knowledge of attacks and vulnerabilities. For example, the Common Vulnerabilities and Exposures (CVE) database records the disclosed vulnerabilities.[4] and the Common Weakness Enumeration (CWE) database lists the weakness in software and hardware[5] In addition, many security reports about cyber attacks are published by many companies and institutions and the knowledge of attack steps and tools can be extracted from these reports. Furthermore, the security forums release news about cyber attacks and defenses, which play an important role in extracting cyber security knowledge.

The cyber security knowledge can be extracted from various data sources and many classical extraction methods can be adopted, such as entity extraction, relation extraction, etc. Following that, this module converts the knowledge to a MDATA node, which consists of a large-scale MDATA graph. There are many ways to represent cyber security knowledge and we introduce our method as follows.

Specifically, we focus on three categories of cyber security knowledge: assets, weaknesses, and attacks. Hence, we extract knowledge from these three aspects. An example is shown in Fig. 3, where the asset type contains a computer and the Adobe Reader software, the weakness type includes a specific vulnerability (CVE-2008-2992), and the attack type has the buffer overflow attack. According to the extracted knowledge, the Adobe Reader software exists a vulnerability if the version meets certain conditions, hence the node adobe reader and vulnerability are connected. Similarly, the buffer overflow attack exploits the

[4] https://cve.mitre.org
[5] https://cwe.mitre.org

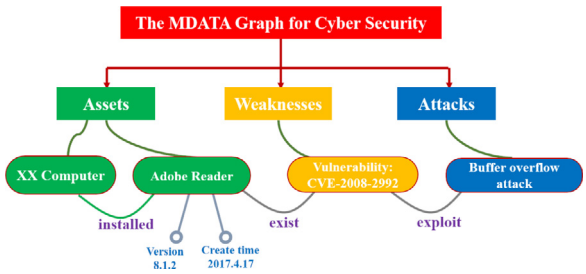

Fig. 3. An example of the MDATA graph for cyber security knowledge.

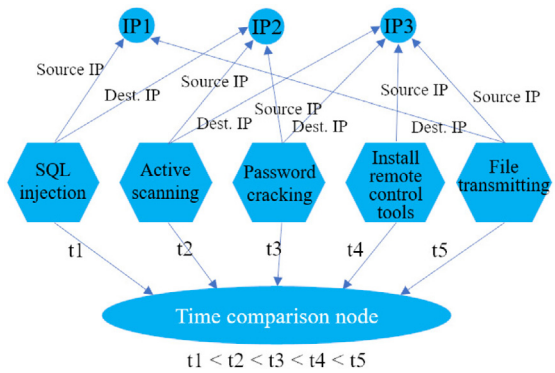

Fig. 4. An example of converting the attack detection rule to the MDATA subgraph.

vulnerability, hence these two nodes are connected. As shown in the figure, the cyber security knowledge can be extracted and represented in the MDATA graph and it is easy to infer the computer might be attacked by the buffer overflow attack. The established MDATA graph plays an important role to assist multi-step attack detection.

4.3. Subgraph generation module

A networked system consists of many assets and attackers aim to obtain important data or files through cyber attacks. Traditional methods deal with the huge number of security alarms produced by security products, the analysis and detection efficiency are greatly impacted since many unrelated or false alarms exist. Hence, this module aims to generate a subgraph related to the networked system, which helps to filter out these false or unrelated alarms.

In detail, the module applies asset scanning and vulnerability scanning tools to generate the assets and the weaknesses lists $A_G = \{A_G(v_1), A_G(v_2), \ldots, A_G(v_n)\}$ and $W_G = \{W_G(v_1), W_G(v_2), \ldots, W_G(v_m)\}$ for the system $G = (V, E)$. Since the module can obtain a detailed list of the assets and the weaknesses, the associated attacks that might be launched can be inferred through the established MDATA graph. Hence, the module generates the subgraph for the system from the MDATA graph for cyber security, which contains the potential attacks on the system. Then, the detection framework can filter out unrelated alarms to improve detection efficiency.

Considering the potential attacks, we also convert the detection rules to a subgraph, which enables the persistent detection capability. As shown in Fig. 4, the detection rule for the multi-step attack in Fig. 2 is converted to a MDATA subgraph. There are three layers in the figure, with the top layer representing the IP addresses involved, the middle layer representing the specific attacks, and the bottom layer representing the temporal relation.

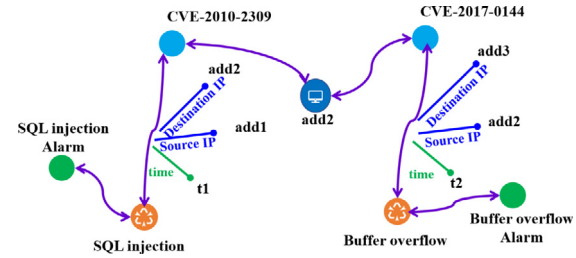

Fig. 5. An example of constructing the MDATA graph for detection through the alarm correlation module.

Each node in the middle layer is connected to the source IP and destination IP (dest. IP for short in the figure), and the timestamp is connected to the time comparison node. As the five attack steps in Fig. 2 should be executed sequentially, the bottom node compares the temporal relations of the five attacks. If these attacks are detected such that their IP characteristics and temporal relations both satisfy the detection graph, then the multi-step can be detected. We can convert all attack detection rules to such MDATA subgraphs and we refer to them as the *detection subgraphs*.

4.4. Alarm correlation module

Normally, different kinds of security products are deployed and the descriptions of the security alarms are different. Hence, this module first fuses these security alarms such that they can be represented by a unified form. After that, the module removes unrelated alarms according to the subgraph generated for the networked system in the subgraph generation module. To ensure the attack capabilities, the detection rules in the security products are very sensitive and many false alarms are generated. Once the subgraph of the system is generated, we can focus on only the potential attacks and thus remove the unrelated alarms. This step helps reduce the number of alarms and can improve detection efficiency.

As the alarms are in a textual form, we need to extract the attack information and convert it to a MDATA graph. Specifically, we need to extract effective information about the alarm flow $ALARM_G = \{alarm_G(t_0), alarm_G(t_1), \ldots\}$ into a MDATA graph. The attack extraction process can identify the source IP IP_s, source port $Port_s$, destination IP IP_d, destination port $Port_d$ and timestamp of the attack. Then, the alarm correlation process connects these alarms to the attacks in the MDATA graph for further detection. For example, the module extracts the SQL injection alarm and connects it to the attack node in Fig. 5. The attack step information is described with destination IP, source IP and time information. The buffer overflow alarm is similarly represented in the MDATA graph. As the security alarms are constantly generated, the MDATA graph for detection grows with more attack information and calls the MDATA graph as *a real-time attack graph*.

4.5. Attack detection module

After the real-time attack graph and the detection subgraphs are generated, the attack detection module aims to detect all multi-step attacks in a short time.

There are some important factors in the module. For example, spatio-temporal indexing can accelerate the process of querying since MDATA model already presented spatial and temporal characteristics. Hence, the multi-hop querying can be finished quickly compared to the existing methods that must associate both spatial and temporal factors. Then, subgraph matching technologies

are adopted in the module to find the subgraphs in the real-time attack graph that can be matched with the detection subgraph. In other words, the rule matching process according to the detection rules can be greatly improved by the subgraph matching technologies. The attack step association process restores the detailed steps of each attack and generates the detection report.

Obviously, many subgraph matching methods can be applied in the module. In this part we also introduce a simple but efficient matching algorithm as Alg. 1. First, we initialize a list to store the matching results and take the continuous alarm logs $ALARM_G = \{alarm_G(t_1), alarm_G(t_2), \ldots\}$ as input. After each alarm $alarm_G(t_i)$ converting to a MDATA node, we query whether it is the first step of any attack or a follow-up step of a previously matched step of the detection subgraph. Notice that the spatial and temporal constraints should be satisfied if it is recognized as a follow-up step. By repeating the process, the algorithm outputs a detected multi-step attack once all the attack steps have been found.

Algorithm 1 Subgraph Matching based Multi-step Attack Detection

Input: The networked system $G = (V, E)$, the asset list A_G, the weakness list W_G, the detection subgraphs $SG_D = \{SG(1), SG(2), \ldots, SG(m)\}$; the real-time alarms $ALARM_G = \{alarm_G(t_0), alarm_G(t_1), \ldots\}$;
Output: The detected multi-step attacks;

```
Initialize list l_match = ∅;
for each alarm alarm_G(t_i) do
    convert it to a MDATA node M_a(t_i);
    if M_a(t_i) is the first step of a graph SG(j) then
        add M_a(t_i) to the jth list in l_match;
    else if M_a(t_i) is a follow-up step of the jth list in l_match, and
    the spatial and temporal constraints are satisfied according to
    SG(j) then
        add M_a(t_i) to the end of the jth list in l_match;
    end if
    if the jth list of l_match consists SG(j) then
        output the multi-step attack of SG(j);
    end if
end for
```

4.6. Advantages and disadvantages

The proposed framework can work persistently to detect multi-step attacks. Compared to the existing attack detection methods, the framework has the following advantages:

- The capability of detecting multi-step attacks. Many existing methods can only find out a specific attack by rule-based or machine learning-based methods and they lack the correlation analysis of multiple security data sources. The proposed framework can associate the security alarms to find out all multi-step attacks;
- The ability to detect valid attacks. The proposed subgraph generation module can generate the subgraph and focus on the potential attacks. This module enables the removal of false alarms and can help to detect the valid attacks that might be launched on the system;
- The efficiency of attack detection. The MDATA model can largely reduce the multi-hop query time by spatial –temporal indexing, which increases the efficiency of attack detection. In addition, the subgraph matching method shows superior performance when dealing with the streaming security alarm data.

Although the framework has significant detection capabilities and high efficiency, there are some disadvantages. First, the detection subgraphs are converted from the attack detection rules. When the attack technologies or steps change, the subgraphs will also change once the detection rules are obtained. Second, the real-time detection graph grows as security alarms are constantly produced, and it is difficult to decide the time window to generate the detection graph. If the window is large, the attack graph will be very large to manage, while some attacks that have been hiding for a long time cannot be detected if the window is small. Therefore, achieving balance is an important and meaningful study topic in the future.

Table 1
Description of three datasets.

Dataset	# of nodes	# of edges
Graph500	2.4 million	67 million
Twitter	41.6 million	1.4 billion
Cyber security	120 million	6.5 billion

5. Experimental results

In this section, we evaluate the proposed framework from three aspects. First, we evaluate the performance of the MDATA model in representing and managing cyber security knowledge. Then, we evaluate the efficiency and accuracy of the proposed attack detection algorithm based on subgraph matching. Finally, we conduct experiments on the cyber range platform to evaluate the performance of the proposed framework.

5.1. Experiment setting

In our experiments, we use 4 high performance computing servers to store and manage cyber security knowledge. In addition, two commonly used datasets Graph500[6] and Twitter user-follower graph[7] are used to evaluate the management capabilities of the MDATA graph. We establish a large networked system on the cyber range platform with 5 kinds of security products and use some pre-designed attack data to evaluate the proposed framework. Specifically, we use 10 high performance computing servers (CPU: 8 cores, GPU: Tesla T4) and 30 virtual computing servers (CPU: 4 cores) to build the virtual networked system.

5.2. Management of cyber security knowledge

Through the knowledge extraction module, we have extracted a large amount of cyber security knowledge and converted them to the MDATA graph, which consists of 1.2 billion nodes. We use the Graph500 and Twitter user-follower graph datasets and a part (about 10%) of the cyber security knowledge. The details of the three datasets are described in Table 1.

In order to evaluate the knowledge management capabilities, we use the multi-hop querying latency to compare it with some classical graph computing systems: TigerGraph, Neo4j, and JanusGraph. As shown in Tables 2 and 3, with introduction of temporal-spatial indexing in the MDATA graph, the MDATA graph can support faster one-hop and two-hop queries. Specifically, the one-hop query time can be reduced to less than 25 ms for a large knowledge base and the two-hop query time is also limited within 670 ms. The multi-hop querying efficiency can accelerate the subgraph matching based multi-step attack detection.

6 http://graph500.org
7 http://an.kaist.ac.kr/traces/WWW2010.htm

Table 2
Comparison of one-hop Query Time (ms).

Dataset	TigerGraph	Neo4j	JanusGraph	MDATA graph
Graph500	6.3	21	27.2	5.1
Twitter	24.1	205	394.8	18
Cyber security	37.5	387.3	712.3	25

Table 3
Comparison of two-hop Query Time (ms).

Dataset	TigerGraph	Neo4j	JanusGraph	MDATA graph
Graph500	310	4180	15390	210
Twitter	460	18340	27780	510
Cyber security	512	27355	39355	670

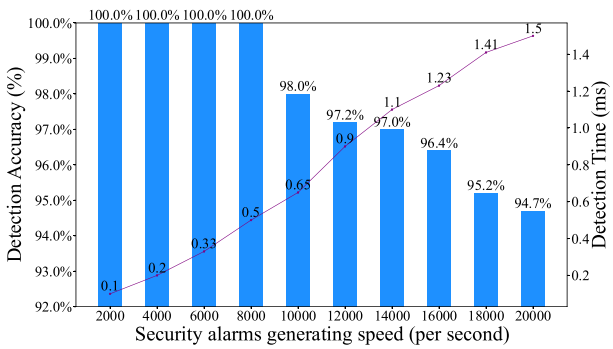

Fig. 6. Detection accuracy and detection time when K increases.

5.3. Subgraph matching based attack detection

We evaluate the subgraph matching based attack detection algorithm (Alg. 1) by generating flow security alarms. As there are many existing multi-step attacks, we select 100 multi-step attacks to generate the corresponding security alarms, including the source IP, source port, destination IP, destination port, timestamp, attack description, and other textual information.

We first evaluate the computation efficiency when a large number of alarms are generated. We select 50 multi-step attacks and generate the corresponding security alarms. Thus, the input detection subgraphs $SG_D = \{SG(1), SG(2), \ldots, SG(50)\}$. We generate K alarms per second, where K increases from 2000 to 20 000 and we generate the alarms during an hour. As shown in Fig. 6, when K reaches 10 000, some multi-step attacks cannot be detected, because the generated alarms are all related to the attacks and the algorithm must store these attacks. When the generation speed is high, the memory is not enough and some attack steps are lost in the algorithm. By increasing the hardware capability, this problem can be solved. From the figure, it is evident that the detection time increases as more alarms are generated because the algorithm has to make more correlations and detection steps.

Then, we evaluate the detection time when the number of detection subgraphs increases. We generate the security alarms for all 100 multi-step attacks with different speeds: 2000,5000 and 10000 per second. We assume the subgraph generation module only finds T potential attacks and the corresponding T detection subgraphs are used in the algorithm. As shown in Fig. 7, the detection time increases as the number of detection subgraphs increases. This is because many unrelated or false alarms are filtered when detecting the attacks. This implies the subgraph generation module can help to remove the false alarms to improve the detection accuracy. From the figure, when the alarm generation speed increases, the detection time also increases, which is also consistent with the phenomenon in Fig. 6.

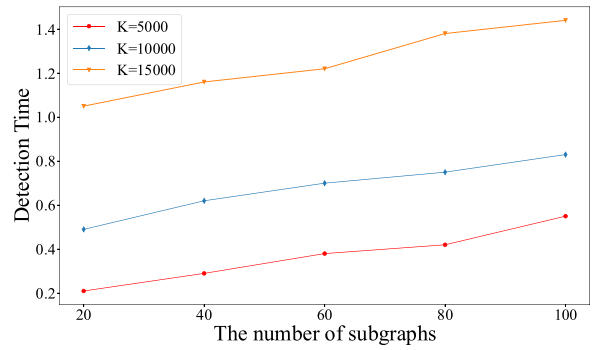

Fig. 7. Detection time when the number of detection subgraphs T increases.

Table 4
Attack detection accuracy and efficiency on the simulated system.

Attack type	Detection accuracy	Detection time
DDoS	99%	20 ms
Active scanning	99%	1 ms
Exploit attack	95%	1 ms
Some known APT	90%	539 ms
All types	90.6%	14 ms

5.4. Framework evaluation on the cyber range platform

We implement the ACAM framework on the cyber range platform, which builds on 10 high performance computing servers and 30 virtual computing servers. The cyber range platform can simulate a virtual networked system, including the nodes, edges, assets, and weaknesses of each node. The cyber range platform can assist various competitions and drill operations and we have collected various kinds of attacks during these activities, including many single-step and multi-step attack data. Then we launch these attacks into the system and evaluate the detection accuracy and efficiency according to the collected data.

As shown in Table 4, the detection accuracy and efficiency of some kinds of attacks are listed. As the cyber range platform can simulate a real networked system, we tested the framework for detecting different kinds of attacks and demonstrated the average detection accuracy and detection time. The average detection accuracy can reach as high as 90.6% and the average delay is relatively low at 14 ms.

6. Conclusions

In this paper, we propose ACAM, an advanced cyber security analysis framework based on the MDATA model, which enables efficient detection of cyber attacks. The framework is composed of four modules and a large-scale MDATA graph for cyber security knowledge. The knowledge extraction module extracts the latest cyber security knowledge from various kinds of data sources and converts the knowledge into MDATA graph to support efficient query and association. The subgraph generation module reduces false alarms and improves detection efficiency. The alarm correlation module establishes the real-time attack graph according to the constantly produced alarms and performs the subgraph matching in the attack detection module, which can facilitate the detection of multi-step attacks. The framework plays an important role in cyber security defense with artificial intelligence abilities. We implemented the framework on the cyber range platform and conducted extensive experiments to evaluate the detection performance. Considering the detection efficiency, the

proposed framework can significantly improve the cyber security defense capabilities in smart cities. As the proposed method is limited by the varied detection rules and the time window for generating the detection graph, we will continue the optimization of the framework and the detection algorithm in the future.

CRediT authorship contribution statement

Yan Jia: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Methodology, Conceptualization. **Zhaoquan Gu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Conceptualization. **Lei Du:** Writing – original draft, Visualization, Software. **Yu Long:** Writing – original draft, Methodology, Data curation. **Ye Wang:** Writing – review & editing, Data curation. **Jianxin Li:** Data curation, Writing – review & editing. **Yanchun Zhang:** Formal analysis, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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